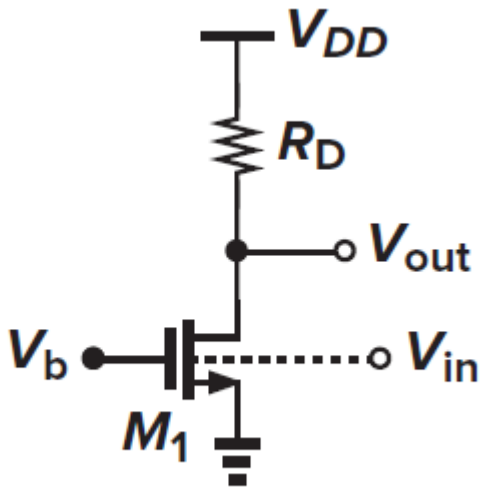
Exercise 4-2
Single-Stage Amplifiers

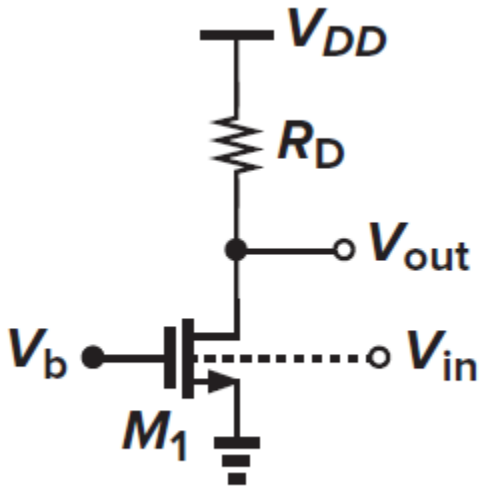
Winter Term 2025/26

Nima Shahpari, Ph.D.

- It is possible to use the bulk (back gate) of a MOSFET as the terminal controlling the channel. Determine the voltage gain if $\lambda = 0$.



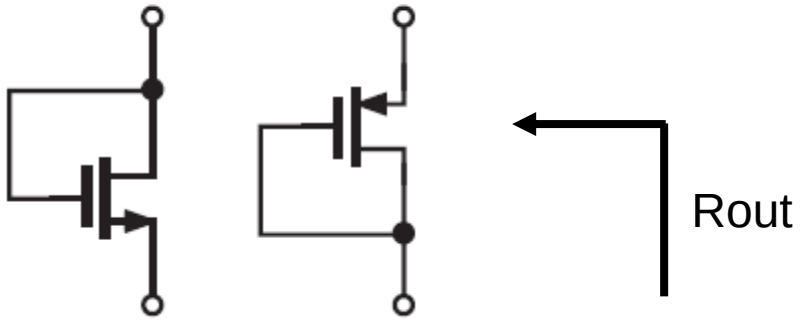
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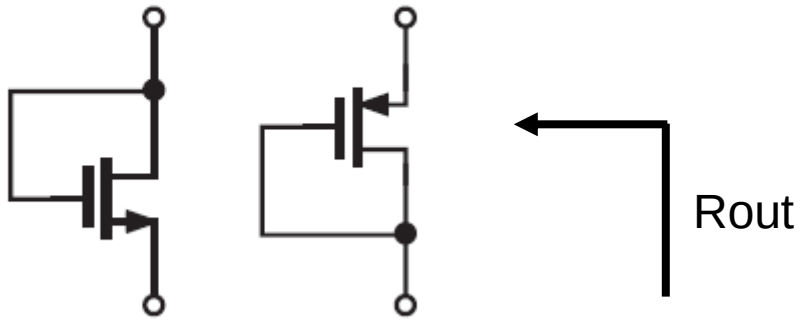
From the small-signal MOS model developed in Chapter 2, we recall that the drain current is given by $g_{mb}V_{in}$.

Thus, $A_v = -g_{mb}R_D$.

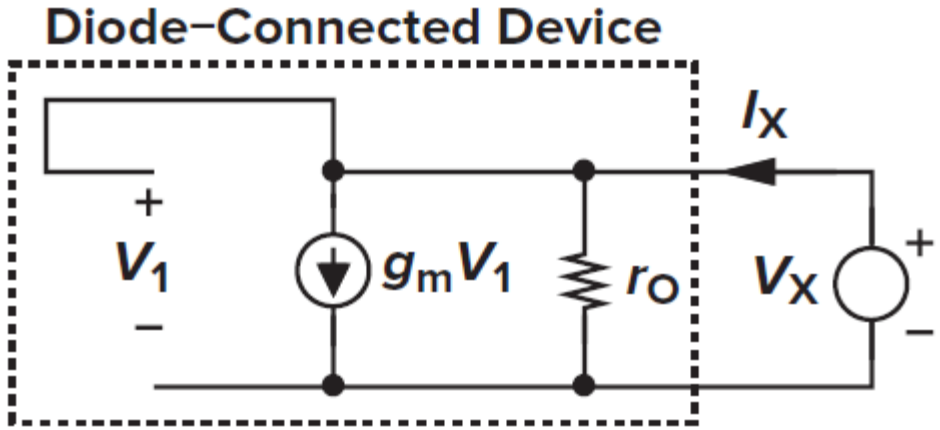
- Calculate the output Resistance in a Diode connected structure.



- Calculate the output Resistance in a Diode connected structure. Ignoring the body effect.



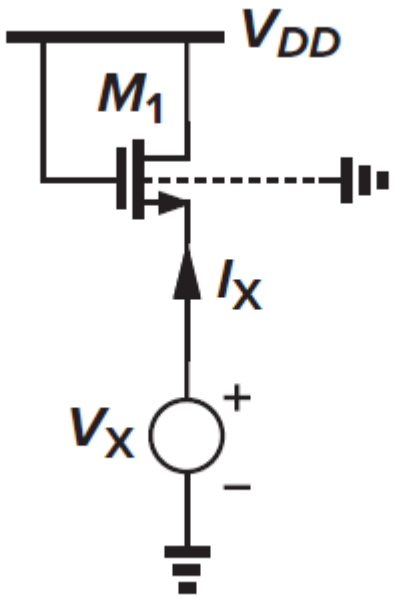
Rout



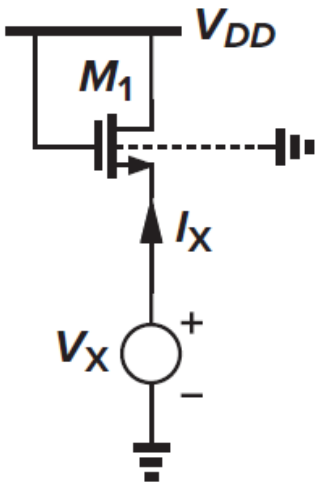
$V_1 = V_X$ *and* $I_X = V_X/r_O + g_m V_X$

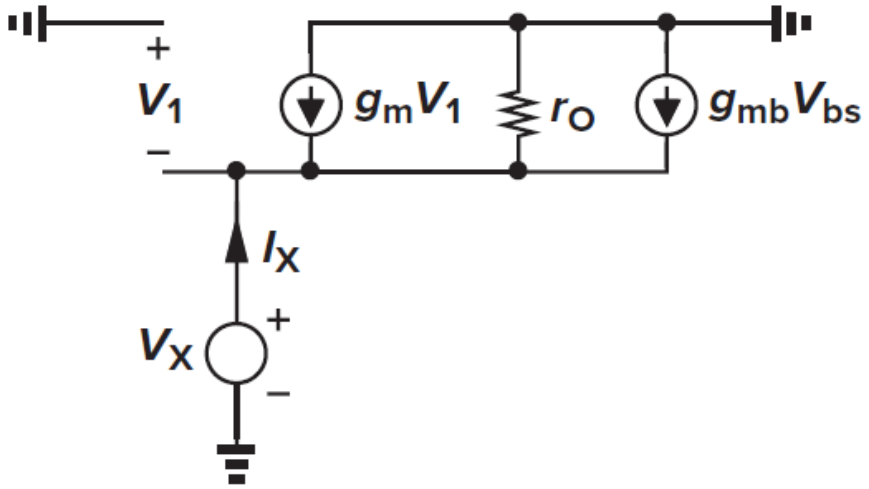
$$Rout = V_X/I_X = (1/g_m) \| r_O \approx 1/g_m$$

- Calculate the output Resistance in a Diode connected structure. consider the body effect.

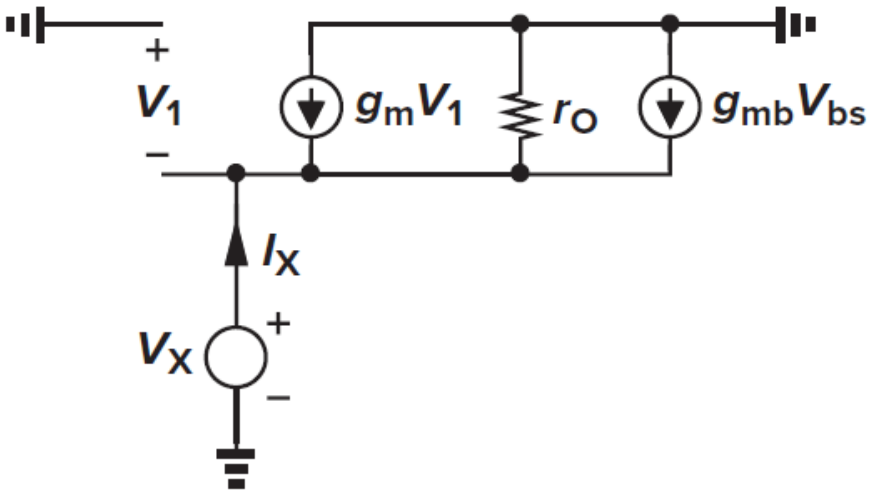


- Calculate the output Resistance in a Diode connected structure. consider the body effect.





- Calculate the output Resistance in a Diode connected structure. consider the body effect.



$$V_1 = -V_X, \qquad V_{bs} = -V_X$$

$$(g_m + g_{mb})V_X + \frac{V_X}{r_O} = I_X$$

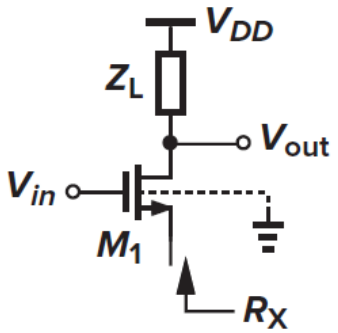
$$\frac{V_X}{I_X} = \frac{1}{g_m + g_{mb} + r_O^{-1}}$$

$$= \frac{1}{g_m + g_{mb}} \| r_O$$

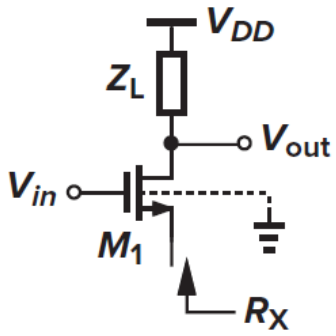
$$\approx \frac{1}{g_m + g_{mb}}$$

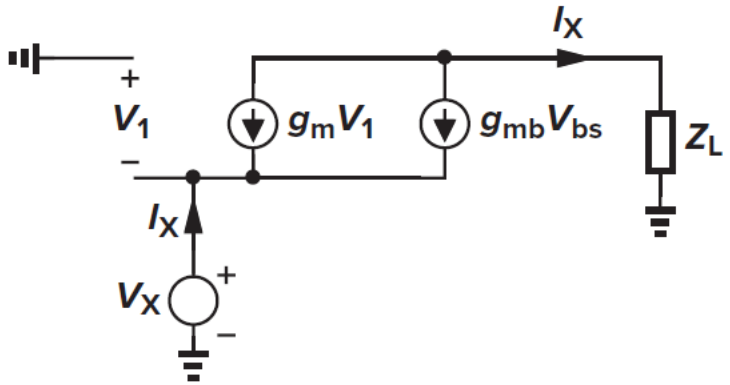
$$V_X/I_X = (1/g_m)||r_O||(1/g_{mb})$$

- Consider the circuit shown in Fig. 3.12(a). In some cases, we are interested in the impedance seen looking into the source, R_X. Determine R_X if $\lambda = 0$.



- Consider the circuit shown in Fig. 3.12(a). In some cases, we are interested in the impedance seen looking into the source, R_X. Determine R_X if $\lambda = 0$.





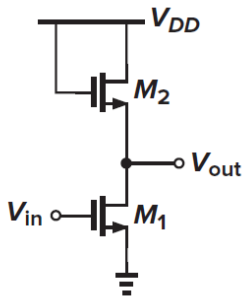
$$V_1 = -V_X \text{ and } V_{bs} = -V_X$$

$$(g_m + g_{mb})V_X = I_X$$

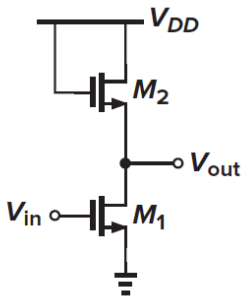
$$\frac{V_X}{I_X} = \frac{1}{g_m + g_{mb}}$$

We sometimes say, "looking into the source of a MOSFET, we see $1/g_m$," assuming implicitly that $\lambda = \gamma = 0$.

- Calculate the gain of a CS with diode connected as a load. Neglect the channel length modulation.



- Calculate the gain of a CS with diode connected as a load. Neglect the channel length modulation.

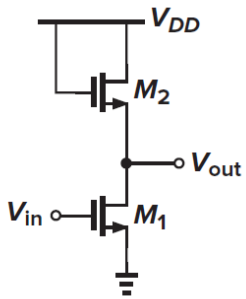


$$A_v = -g_m R_{Out} \qquad A_v = -g_{m1}\frac{1}{g_{m2}+g_{mb2}} = -\frac{g_{m1}}{g_{m2}}\frac{1}{1+\eta} \qquad \text{where } \eta = g_{mb2}/g_{m2}$$

$$A_v = -\sqrt{\frac{(W/L)_1}{(W/L)_2}}\frac{1}{1+\eta}$$

This equation reveals an interesting property: if the variation of η with the output voltage is neglected, the gain is independent of the bias currents and voltages.

- Plot the Vout-Vin plot for a CS with Diode connected load.



- Plot the Vout-Vin plot for a CS with Diode connected load.

